Q.23 Explain elasticity and plasticity.

Q.24 Differentiate crystalline and amorphous materials.

Q.25 Explain Gibbs phase rule.

Q.26 Write properties of clays.

Q.27 Explain creep behaviour of materials.

Q.28 Define thermal expansion.

Q.29 Explain conductivity and resistivity.

Q.30 Write applications of phase rule.

Q.31 Explain structures of silica.

Q.32 Name magnetic properties of materials.

Q.33 Write types of crystal systems.

Q.34 Explain water system phase diagram.

Q.35 Write short note on casting slips.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain soda silica phase diagram.

Q.37 Describe electrical and thermal properties of materials.

Q.38 Discuss in detail physical nature of clay.

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3rd Sem / Ceramic Engineering
Sub : Ceramic Science

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The number of valence electrons in Potassium atoms are

a) 2 b) 1

c) 3 d) 4

Q.2 Which bond is strongest?

a) Hydrogen bond b) Ionic bond

c) Covalent bond d) Metallic bond

Q.3 The number of components in binary system is

a) 1 b) 2

c) 3 d) 4

Q.4 Example of point defect is

a) Dislocation b) Vacancy

c) Crack d) Grain boundary

Q.5 Hard magnetic materials have

a) Low coercivity b) High coercivity

c) No magnetism d) Zero permeability

Q.6 Phase rule is expressed as

a) $F = C - P + 2$ b) $F = P - C + 2$

c) $F = C + P - 2$ d) $F = C - P - 2$

Q.7 Clay becomes plastic due to

a) Heat b) Water

c) Pressure d) Cooling

Q.8 Electrical resistance is measured in

a) Ohm b) Volt

c) Ampere d) Watt

Q.9 Crystalline materials have

a) Random structure

b) Regular structure

c) No structure

d) Liquid structure

Q.10 Example of ternary phase diagram is

a) Water system b) Na_2O-SiO_2-CaO

c) Alumina-Silica d) Oxygen system

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Maximum no of electrons in d-orbital is _____.

Q.12 Covalent bond is formed by sharing of electrons. (True / False)

Q.13 The formula of kaolin is _____.

Q.14 Uniary phase system contains _____ components.

Q.15 Crystalline solids show long range order. (True / False)

Q.16 Magnetic flux is measured in _______.

Q.17 Plasticity is property of clay. (True / False)

Q.18 Thermal stress is caused due to _____.

Q.19 Phase rule was given by ______.

Q.20 Fatigue failure occurs due to _____ loading.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain atomic structure of atom.

Q.22 Define ionic bond with example.